

Real-Time SAR Ship Detection: A Speckle-Model-Based DSP Pipeline

Date: 2026-08-24

Presented by: Jen-Tse Chen

Advisor: Pei-Yun Tsai

Outline

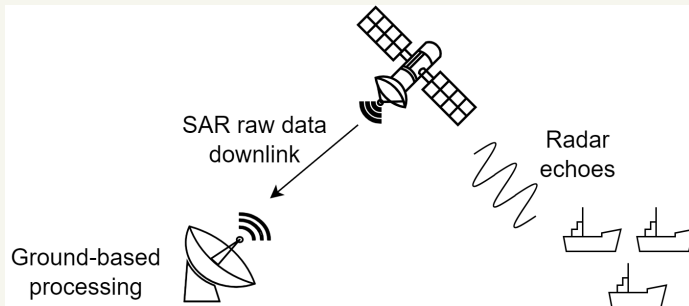
- ▶ Introduction and Motivation
- ▶ Evaluation Framework
- ▶ SAR Echo and Imaging Model
- ▶ Sparse Image Refinement
- ▶ Experimental Results
- ▶ Conclusion
- ▶ Reference

Outline

- ▶ Introduction and Motivation
- ▶ Evaluation Framework
- ▶ SAR Echo and Imaging Model
- ▶ Sparse Image Refinement
- ▶ Experimental Results
- ▶ Conclusion
- ▶ Reference

Introduction

- ▶ SAR raw data are downlinked for ground-based processing, including SAR focusing, image enhancement, and ship detection.
- ▶ Detected ships provide information for maritime surveillance.



Problem and Motivation

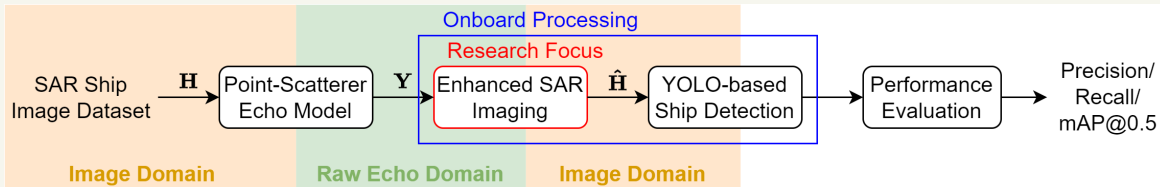
- ▶ Ground-Based Processing
 - ▶ Large raw-data volume
 - ▶ Limited downlink bandwidth
 - ▶ Delayed target information
- ▶ Onboard Processing
 - ▶ Reduced data transmission and detection latency
 - ▶ Challenge: Limited computation, memory, and power
- ▶ Research Goal
 - ▶ Lightweight SAR processing for low-latency ship detection

Outline

- ▶ Introduction and Motivation
- ▶ Evaluation Framework
- ▶ SAR Echo and Imaging Model
- ▶ Sparse Image Refinement
- ▶ Experimental Results
- ▶ Conclusion
- ▶ Reference

Evaluation Framework

- Research Focus: Enhanced SAR imaging
- Evaluation Task: YOLO-based ship detection



► Evaluation Metrics

- Precision: $\frac{TP}{TP+FP}$
- Recall: $\frac{TP}{TP+FN}$
- mAP@0.5: Mean AP across classes at IoU threshold = 0.5

TP: True Positive, FP: False Positive, FN: False Negative

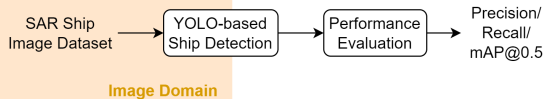
Dataset and Detection Model Setup

► HRSID

- High-Resolution SAR Images Dataset
- Training: 3,642 images
- Validation: 1,962 images
- Image size: 800×800 pixels

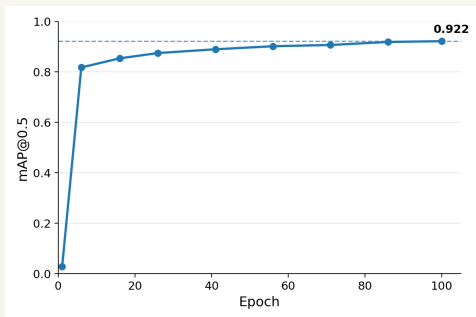
► YOLO Detection Model

- YOLO11m fine-tuned on HRSID (100 epochs)



► Validation Performance

- Precision: 0.910
- Recall: 0.840
- mAP@0.5: 0.922

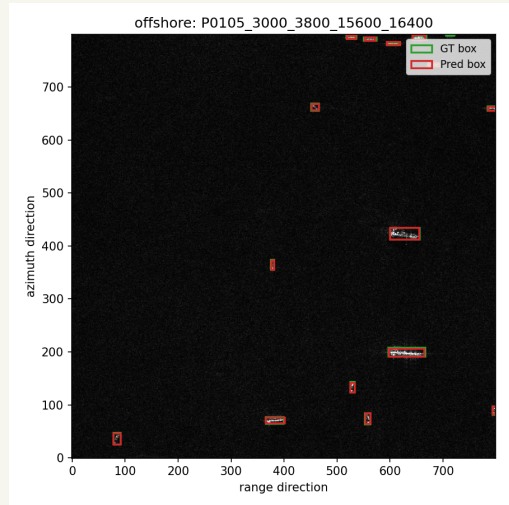


YOLO Ship Detection Results (1)

	Overall	Inshore	Offshore
Images	1,962	369	1,593
Precision	0.913	0.803	0.982
Recall	0.846	0.741	0.965
mAP@0.5	0.918	0.817	0.984

- Offshore scenes achieve better detection performance than inshore scenes.

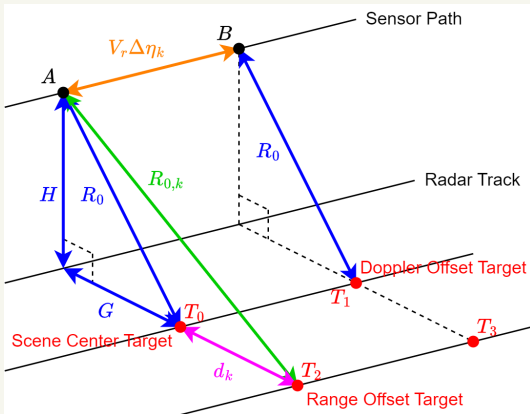
YOLO Ship Detection Results (2)



Outline

- ▶ Introduction and Motivation
- ▶ Evaluation Framework
- ▶ SAR Echo and Imaging Model
- ▶ Sparse Image Refinement
- ▶ Experimental Results
- ▶ Conclusion
- ▶ Reference

Point-Scatterer Echo Model: Geometry



- Target position determines the time-varying slant range $R_k(\eta_m)$.
- $$R_k(\eta) = \sqrt{R_{0,k}^2 + (\eta_m + \Delta \eta_k)^2 V_r^2}$$
 - $R_{0,k} \approx R_0 + d_k$
 - R_0 : slant range to scene center
 - d_k : ground-range offset of scatterer k , approximated as slant-range offset
 - $\eta_m = m \Delta \eta$: slow time
 - $\Delta \eta_k$: azimuth-time offset of scatterer k

Point-Scatterer Echo Model: Signal Synthesis

► Geometry-Dependent Delay and Phase

► $\tau_k(\eta_m) = \frac{2R_k(\eta_m)}{c}, \phi_k(\eta_m) = -4\pi \frac{R_k(\eta_m)}{\lambda}$

► Echo Synthesis: $\mathbf{S} = [s_{m,n}]_{M \times N}$

$$s_{m,n} = \sum_k A_k \text{rect}\left(\frac{\tau_n - \tau_k(\eta_m)}{T_r}\right) \cdot e^{j\pi K_r (\tau_n - \tau_k(\eta_m))^2} \cdot e^{j\phi_k(\eta_m)}$$

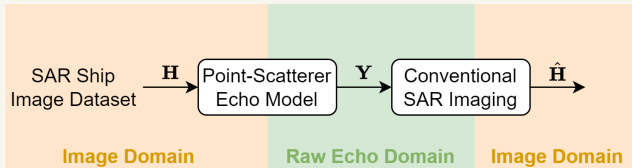
► $\eta_m = m\Delta\eta$: slow time, $\tau_n = n\Delta\tau$: fast time

► A_k : scattering coefficient of scatterer k , $K_r = \frac{B_r}{T_r}$

Outline

- ▶ Introduction and Motivation
- ▶ Evaluation Framework
- ▶ SAR Echo and Imaging Model
- ▶ Sparse Image Refinement
- ▶ Experimental Results
- ▶ Conclusion
- ▶ Reference

Conventional SAR Imaging



- Imaging goal: Recover $\mathbf{H}$ from $\mathbf{Y}$.
- Conventional methods: RDA, CSA, and BP

One-Step Sparse Image Refinement

► Image-domain regularization

$$\hat{\mathbf{H}} = \arg \min_{\mathbf{H}} \underbrace{\frac{1}{2} \|\mathbf{H} - \mathbf{H}_{\text{CSA}}\|_F^2}_{\text{Preserve CSA image}} + \lambda \underbrace{\|\mathbf{H}\|_1}_{\text{Promote sparsity}}$$

► Strong scatterers $\rightarrow$ retained; weak components $\rightarrow$ suppressed

► $\hat{\mathbf{H}} = \mathcal{S}_{\lambda}(\mathbf{H}_{\text{CSA}})$, $\mathcal{S}_{\lambda}(x) = \frac{x}{|x|} \max(|x| - \lambda, 0)$, $x \neq 0$

► One-Step Refinement

► $\mathbf{Y} \xrightarrow{\text{CSA}} \mathbf{H}_{\text{CSA}} \xrightarrow{\mathcal{S}_{\lambda}(\cdot)} \hat{\mathbf{H}}$

Outline

- ▶ Introduction and Motivation
- ▶ Evaluation Framework
- ▶ SAR Echo and Imaging Model
- ▶ Sparse Image Refinement
- ▶ **Experimental Results**
- ▶ Conclusion
- ▶ Reference

Simulation Setup

- Dataset: 100 offshore SAR images
- Evaluation: Ship detection on reconstructed SAR images
- SAR Simulation Parameters

Parameter	Value	Parameter	Value
Wavelength	0.1152 m	Closest Slant Range	4000 m
Pulse Width	12.5 μ s	Height	0 m
Pulse Repetition Frequency	1 kHz	Platform Velocity	120 m/s
Bandwidth	50 MHz	Antenna Length	1.2 m
Sampling Frequency	64 MHz		

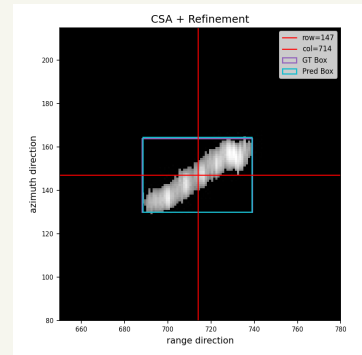
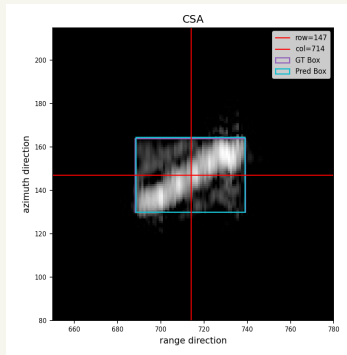
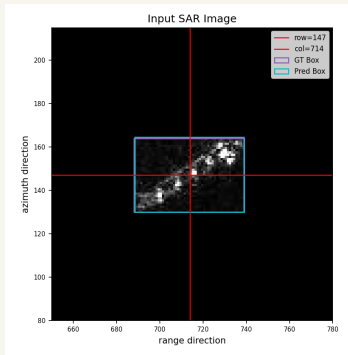
Ship Detection Performance

Image	Precision	Recall	mAP@0.5
Input SAR Image	1.000	0.978	0.975
CSA	0.977	0.940	0.954
CSA + Refinement	0.993	0.986	0.985

- Thresholding suppresses CSA-induced background artifacts and improves ship detection performance.

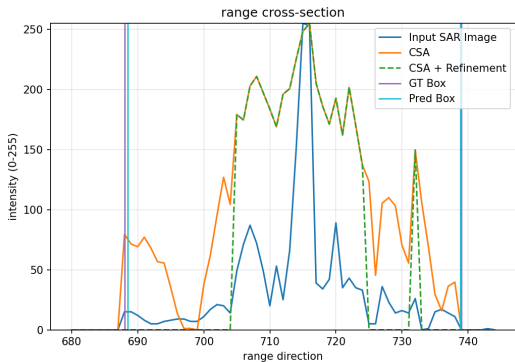
Representative Example

- Observation: Thresholding suppresses background artifacts while preserving the dominant ship response.



Cross-Section Analysis

- Thresholding removes low-amplitude sidelobe/background responses while retaining the dominant target structure.



Outline

- ▶ Introduction and Motivation
- ▶ Evaluation Framework
- ▶ SAR Echo and Imaging Model
- ▶ Sparse Image Refinement
- ▶ Experimental Results
- ▶ Conclusion
- ▶ Reference

Conclusion

- ▶ Established an end-to-end framework for evaluating SAR image enhancement through ship detection.
- ▶ One-step sparse refinement suppresses background artifacts while preserving dominant ship responses.
- ▶ CSA + refinement improves detection performance from 0.954 to 0.985 mAP@0.5.
- ▶ Next step: Extend the refinement toward a lightweight DSP-based onboard SAR processing pipeline.

Outline

- ▶ Introduction and Motivation
- ▶ Evaluation Framework
- ▶ SAR Echo and Imaging Model
- ▶ Sparse Image Refinement
- ▶ Experimental Results
- ▶ Conclusion
- ▶ Reference

Reference